

Quiz 2 - Computational Physics I

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SCORE: 18/20

Date: Thursday 27 March 2025 Duration: 45 minutes

Credits: 20 points (5 questions) Type of evaluation: LAB

Provide short and concise answers to the following items:

1. (4 points) Systems of linear equations

(a) Explain how the Gauss elimination method for solving systems of linear equations works.

-0.5 (b) List the main steps for solving such systems via symbolic algebra with SymPy in Python.

a) We define the augmented matrix (coefficients + independent vector) and start doing row operations until in the last row we have zeros before the coefficient of the last variable. Thus, we have found one of the variables and can use it to compute the others.

b) 1. Declare the variables as symbols.
2. Define the coefficients matrix and the independent vector } how?
3. Use one of the solvers giving it as inputs the coefficients array, the independent vector and the variables. } which one? linsolve?

2. (4 points) Systems of nonlinear equations

(a) Indicate 2 methods that we can use to solve systems of ^{non-linear} linear equations in Python.

(b) Briefly explain how each method works.

① Graphic method (if the equations just involve x, y)

1. Plot each equation as function of y .
2. Calculate the sign of the operation $(y_1 - y_2)$. Then calculate the differences of the signs (use `numpy.diff`). Finally, get the indexes of the non-zero values (using `np.where`).
3. Use the indexes to find the solutions in x and y . } anywhere.

② SymPy (very powerful solvers)?

1. Declare the variables as symbols.
2. Define the coefficients array and the independent vector.
3. Use one of the solvers (nonlinear solve, I'm not sure) with the inputs of before.

3. (4 points) Scientific Data Formats

(a) List 2 types of scientific data formats used in physics.

(b) Briefly describe the features of each data format.

① VTK files

- * Widely used in almost every area of physics.
- * It's a self-describing file. Things like arrays, scalars, units, cell size, etc are all saved.
- * They can store very large data.
- * The cells in the grid can have different sizes, which is very useful in fluid dynamics to correctly simulate the flow around not symmetric objects.

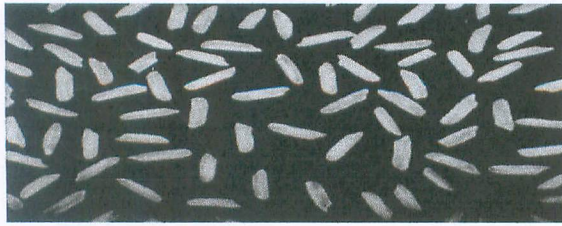
② FITS files

- * Mainly used in astrophysics.
- * It's also a self-describing file.

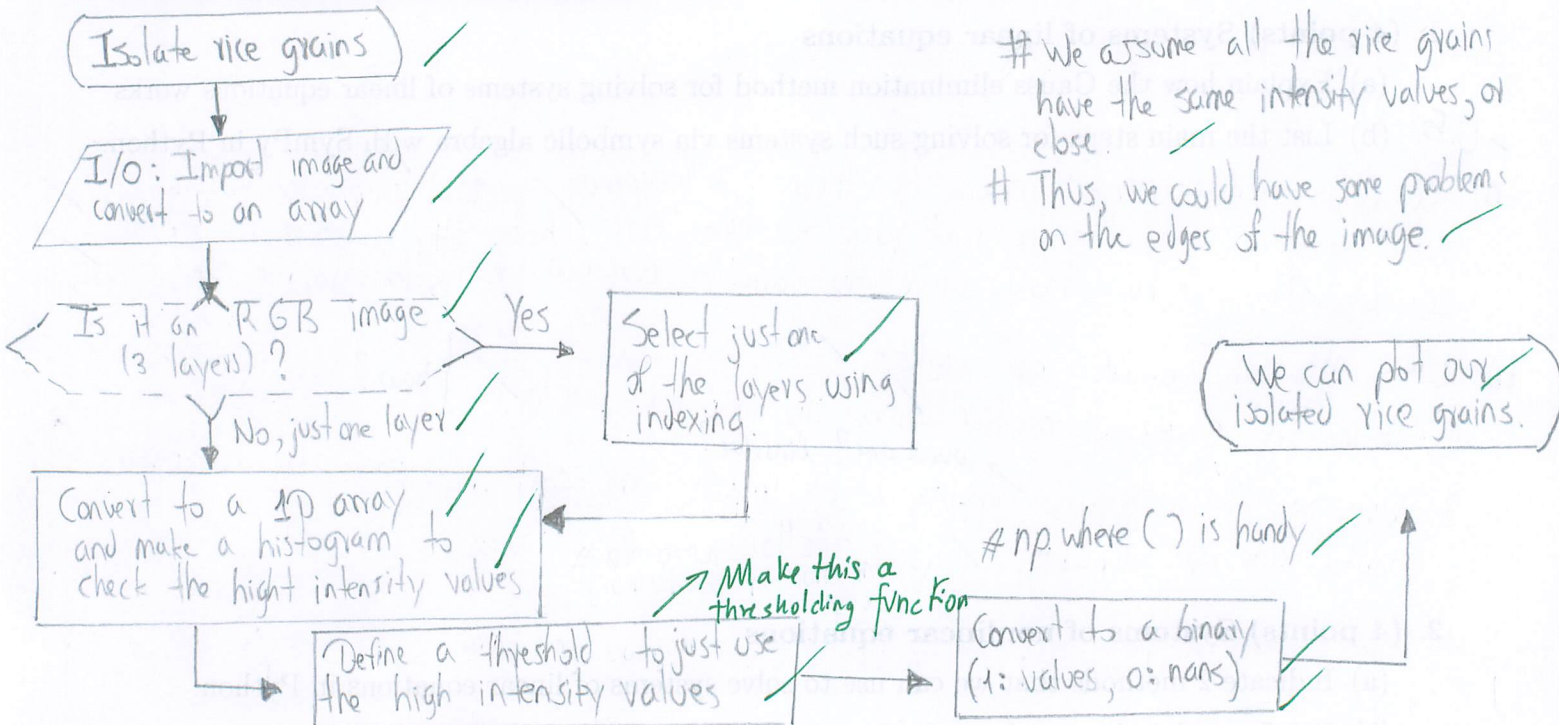
NOTE: Both types of files can be converted to python objects (arrays) and we can work with them more easily.

4. (4 points) Image processing

-0.5



Imagine you obtain the photograph (shown on the left-hand side) of rice, and you are asked to isolate the rice grains from the background. Design and sketch a suitable algorithm workflow to achieve this goal in Python.



5. (4 points) Data processing

You have a dataset consisting of 50 HDF5 files, each containing time, gas density (ρ), and velocity components (v_x, v_y, v_z). Each file corresponds to a different time in the simulation.

- (a) Write a Python function prototype for reading a single HDF5 file, computing the mean kinetic energy, and returning the time and the mean kinetic energy as NumPy arrays.
- (b) Sketch the loop needed to process all 50 files using the above function, so that you obtain and plot the mean kinetic energy versus time.

a) `def mean_k(filename):`
`""" docstring missing`
`file = # I don't recall the method to read a HDF5 file`
`# Comment code`
`t = np.array(file["time"])`
`rho = np.array(file["rho"])`
`vx = np.array(file["vx"])`
`vy = np.array(file["vy"])`
`vz = np.array(file["vz"])`
`# Function prototypes should be commented.`

b) `mk = []`
`for i in len(t):`
 `K = rho[i] * (vx[i]**2 + vy[i]**2 + vz[i]**2) * V`
 `mk.append(K)`
`mk = np.array(mk)`
`return t, mk` `# at time t[i] we compute the mean kinetic energy mk[i].`

Handwritten notes:

- Use vectorisation.
- # V is the volume, can be assumed constant